Slip Face Detection and Route Planning for Small Rovers on Sand Dunes, Tested Against Airborne Lidar

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***Abstract*—Small wheeled robots that cross sand dunes usually plan on a global 30 m elevation model, but the main hazard on active dunes, the slip face, is often only metres long. We built a benchmark on six North American dune fields, with one metre airborne lidar as truth, to measure what coarse maps hide and what a planner can do about it. Ten families of hazard layers, from slope and interpolation to boosted trees, a convolutional network, learned super resolution and imagery fusion, were scored against a lidar face map on lidar block means from 2 to 60 m, on three real elevation products and with Sentinel-2 and NAIP imagery, and were then used by a slip aware planner at operating points chosen on held out sites. At 10 m the best layer raises mission success from 17 % to 64 %. At 30 m no layer, including imagery fusion, adds more than +5 points to 8 %, because most faces are shorter than one cell; a face map from fine data would reach 81 % on the same grid if placed within about 5 m, and onboard sensing within 20 m keeps the rover off faces whatever the prior, although some pairs then have no way through. Code, results and the prediction log are in a public repository.**

***Index Terms*—field robotics, traversability, digital elevation models, sand dunes, slip faces, path planning, benchmark**

I. Introduction

Small wheeled ground robots are proposed for survey, inspection and search in sand deserts, including the dune seas of the Arabian Peninsula. Before such a robot drives, its route is planned on a prior map, and for most deserts the only elevation model available is a global product with a posting of about 30 m [1]. The dominant hazard on active dunes is the slip face, the lee side of a dune held near the angle of repose of dry sand, about 30 to 34 degrees [2], where wheel slip rises until a small vehicle can no longer climb [3], [4]. Slip faces are often only metres to tens of metres long, so a 30 m cell can hold one while its mean slope is gentle.

That a coarse grid understates slope is long known [5]. What is not known is how much this matters for a planner, which hazard layer a practitioner should compute from a coarse map, how fine a map must be, and whether imagery, a different product or onboard sensing change the answer. This paper answers these questions with a benchmark on six North American dune fields where one metre airborne lidar provides the truth.

The contributions are: (1) a benchmark that scores hazard layers against a lidar face map whose meaning is the same at every resolution, on idealised surfaces from 2 to 60 m, on three real elevation products on their own grids, and with Sentinel-2 and NAIP imagery; (2) a comparison of ten families of hazard layers, from slope and interpolation to boosted trees, a convolutional network, learned super resolution and imagery fusion, with every learned layer trained leave one site out; (3) a planning evaluation in which every layer is used at the operating point chosen on the other sites, routes are scored on the lidar, the face map oracle is degraded by position errors and simulated surveys, and a closed loop adds onboard sensing; and (4) a scaling result, detection skill against face length over grid spacing, that indicates which survey resolution a given dune field needs. Code, result files and the prediction log are in the repository listed at the end.

II. Related Work

Overhead priors. Vandapel et al. [6] planned ground vehicle routes on airborne lidar collected before the mission, and overhead imagery or lidar at metre scale has served as a cost prior since [7]. Recent work sends a drone ahead to map only what the ground robot needs [8] or learns terrain properties from a hovering drone [9]. Planetary missions plan from orbital data: traversability analysis selected the Mars 2020 landing site [10], and slip predicted from orbital data drives route plans [11]. These priors are at metre scale; we ask what a 10 to 60 m prior, the only one available for most deserts, hides.

Slip on sand. Geometric costmaps score slope and roughness on a grid [12], [13], [14], and recent reviews cover traversability risk [15]. On loose sand slip rises steeply with slope [3], [16]; Curiosity met high slip crossing megaripples [4], and robophysical studies show how slope and granular flow interact [17]. Planners propagate slip uncertainty into routes [18], [19], [20], but they assume that the map resolves the slope, which fails on dunes at 30 m.

Dunes in elevation models. Coarser grids understate slopes [5]. Dune morphometry from DEMs and imagery is well developed [21], and deep learning maps crestlines from DEM derivatives [22]. Global 30 m products have been ranked against lidar and laser altimetry [23]; TanDEM-X, the source of GLO-30, performs worse over sand [24], and the error of open DEMs in a dune sea grows on slopes above 15 degrees [25]. Learned super resolution can sharpen coarse DEMs [26], [27]. We know of no benchmark that scores coarse map hazard layers and planners for slip faces against lidar truth across resolutions.

TABLE I
LIDAR WINDOWS (USGS 3DEP ONE METRE DEM; FLOWN: ACQUISITION PERIOD OF THE LIDAR PROJECT) AND THE SHARE OF CELLS HOLDING A FACE PATCH POST AT 10 AND 30 M.

Site	Lat., lon.	km^2	Relief	Flown	10 m	30 m
Dumont Dunes, CA	35.65, -116.27	14.4	322 m	2024 to 25	36 %	49 %
White Sands, NM	32.82, -106.29	6.3	20 m	2020	15 %	40 %
Kelso Dunes, CA	34.91, -115.72	9.7	240 m	2024 to 25	15 %	38 %
Great Sand Dunes, CO	37.75, -105.54	14.8	246 m	2019 to 20	16 %	32 %
Algodones Dunes, CA	32.92, -115.10	20.0	122 m	2021 to 23	10 %	27 %
Little Sahara, UT	39.70, -112.39	6.4	99 m	2020 to 21	11 %	35 %

III. BENCHMARK

A. Sites, truth and label

Six windows of USGS 3DEP one metre bare earth lidar [28] cover active dune fields (Table I); we chose them from 20 m slope images and used them in earlier tests of the detector. A post is *steep* when its Horn slope [29] at 1 m is 28° or more, a little below the angle of repose of dry sand [2]; voids never count as steep. A *face patch* is an 8 connected group of at least 10 steep posts (10 m^2). Face patches are steep ground, not verified slip faces. For a coarse cell c with footprint F_c the label is the face map

$$y_c = \mathbb{1}\left[\,\exists\, x \in F_c : x \in \text{a face patch}\,\right], \tag{1}$$

which a planner needs and whose meaning does not change with the cell size. Our earlier label, at least 10 steep posts in F_c, means 10 % of a 10 m cell but 1.1 % of a 30 m cell and is reported only for comparison.

B. Inputs

Idealised: block means of the lidar at 2, 5, 10, 15, 20, 30 and 60 m (I10 and I30 in the tables), which isolate resolution from product noise, time and registration. *Real:* the USGS 1/3 arc second DEM (R10, about 8×10 m) [30], NASADEM (N30, February 2000) [31] and Copernicus GLO-30 (G30, 2011 to 2015) [1], each scored on its own geographic grid, with labels from the lidar posts inside each native cell. Over these sites R10 is probably resampled by USGS from the same lidar collections. *Imagery:* one Sentinel-2 L2A scene per site [32] (10 m; lidar year, cloud below 5 %, lowest sun, chosen by rule) and NAIP [33] (0.3 to 0.6 m, resampled to 1 m).

C. Hazard layers

Table II lists the layers: what a practitioner would compute from a coarse map, learned models, imagery fusion, and a logistic subgrid detector trained only on synthetic dune profiles, which we built earlier for the planner and treat here as one baseline among others. Every learned layer is trained leave one site out [34], so the scored site is never seen in training.

D. Planner

The planner searches the 8 connected grid of the input. For an edge with along track grade $\tan\alpha$ (uphill positive) and cross slope $\tan\beta$,

$$d = \sqrt{(\tan\alpha + f_r)^2 + \tan^2\beta}, \qquad i = -i_0 \ln\left(1 - d/\tan\phi\right) \tag{2}$$

give the traction demand and the slip ($i = \infty$ when $d \geq \tan\phi$), with friction angle ϕ, motion resistance coefficient f_r and slip scale i_0. An edge is passable when $i \leq i_{\max}$ and costs $\ell/(1-i)$ for length ℓ. The four parameters are drawn 128 times; a cell is passable for a draw when at least four of its eight headings are, and q_c is the share of draws for which it is. A cell is BLOCKED when $q_c \leq 0.05$, TRAVERSABLE when $q_c \geq 0.95$ and its face probability $p_c < \tau$, and UNKNOWN otherwise. The main vehicle is a posterior fitted to published slip curves of other wheeled vehicles (ϕ 30.6 to 43.7°, f_r 0.03 to 0.32, i_0 0.33 to 1.21, $i_{\max}$ 0.4 to 0.7; climbing limit 15.9°); a narrower assumed range (limit 25.5°) is the sensitivity. Three policies share the graph: *slip only*, *gate* (TRAVERSABLE cells only) and *weight* (no BLOCKED cell, cost times $1 + w\,p_c$ of the cell entered); gate and weight refuse descents of 28° or more. Every layer is run at $\tau \in \{0.02, \ldots, 0.9\}$ and $w \in \{1, \ldots, 100\}$ and reported at the setting that maximises mission success on the other five sites.

E. Scoring and closed loop

One hundred pairs per site, at least 1 km apart, are drawn twice: E1 from cells whose lidar posts are all gentle (truth used, as in our earlier runs) and E2 from cells whose coarse slope is below 15° (no truth). A route, the polyline through its cell centres sampled every 0.5 m on the lidar, is *unsafe* when a sample falls on a face patch, and a *mission success* when it exists and is not unsafe. In the closed loop the rover follows the plan on a 5 m grid that inherits the coarse classes, costs and face probabilities ($w = 5$); at every step it senses the face patches within a radius R from the lidar, blocks those cells, drops the face penalty of sensed free cells and replans when its route is blocked.

F. Statistics and prediction log

AUC intervals come from a spatial block bootstrap [35] with 300 m blocks; route outcomes carry two stage bootstrap intervals (sites, then pairs). Site level claims use an exact sign flip test over the six sites, Holm corrected [36]; the smallest attainable two sided p is 1/32. On one processor core, assessing a 10 km^2 window and planning 100 routes take 0.2 and 0.2 s at 30 m and 2.3 and 2.2 s at 10 m. Designs and numeric predictions were committed to a prediction log before each run; the log, with 31 hits and 12 misses so far, and the amendments made before runs are in the supplement.

IV. RESULTS

A. Which layer sees faces

Table III gives the mean AUC of every layer for the face map label. On 10 m block means most layers rank face cells

TABLE II

HAZARD LAYERS. L: TRAINED ON LIDAR LABELS OF THE OTHER FIVE SITES (LEAVE ONE SITE OUT). EVERY LAYER THAT IS NOT A PROBABILITY IS CALIBRATED BY LOGISTIC REGRESSION [37] ON THE OTHER FIVE SITES WHEN A PLANNER NEEDS ONE.

Layer	Score of a cell	L
Slope	Horn slope of the coarse surface, the input of a slope costmap	
Ruggedness	Terrain ruggedness index [38] over the 8 neighbours	
Roughness	RMS residual of a plane fitted to 3×3 cells, as in GESTALT [14]	
Aspect rule	Slope times agreement of the cell's downslope azimuth with the dominant face direction of the site (from the coarse map)	
Interpolation	Steepest slope after cubic spline interpolation of the coarse surface to 1 m	
Super resolution	Steepest slope of a 5 m surface predicted by a dilated residual network from the interpolated 30 m surface	yes
Detector	Logistic model on slope, convexity, slope break and relief, trained on 60 000 synthetic straight crested dune profiles sampled at the cell size	
Boosting	Gradient boosted trees [39] on 14 terrain features (slope, ruggedness, roughness, curvature, topographic position, relief, detector features); variants with 5 more features (Boosting+), tuned by nested leave one site out, and without the detector features	yes
Network	Fully convolutional network [40], four 3×3 layers dilated by 1, 2, 4 and 8 [41], 1500 or 6000 steps	yes
+ imagery	Boosting on the terrain features plus Sentinel-2 or NAIP features (band means, brightness, contrast, shadow share); also imagery alone	yes

TABLE III

DETECTION OF THE FACE MAP LABEL (1): MEAN AUC OVER THE SIX SITES. I: LIDAR BLOCK MEANS; I10→30: THE LARGEST OF THE NINE 10 M SCORES IN EACH 30 M CELL; R10: USGS 1/3 ARC SECOND DEM; N30: NASADEM; G30: GLO-30; R10, N30 AND G30 ON THEIR OWN GRIDS. *TRAINED ON LIDAR LABELS OF THE OTHER FIVE SITES. BEST PER COLUMN IN BOLD; · NOT RUN.

Layer	I10	I30	I10→30	R10	N30	G30
Slope	0.90	0.69	·	0.90	0.62	0.65
Ruggedness	0.94	0.76	·	0.93	0.63	0.69
Roughness	0.80	0.77	·	·	·	·
Aspect rule	0.83	0.61	·	·	·	·
Interpolation	0.97	0.82	0.96	**0.96**	0.62	0.70
Super resolution*	·	0.84	·	·	·	·
Detector	0.95	0.74	0.95	0.95	0.56	0.65
Boosting*	0.97	0.81	**0.96**	0.95	**0.63**	**0.72**
Boosting+*	**0.97**	0.83	·	·	·	·
Boosting, tuned*	0.97	0.81	·	·	·	·
Network*	0.96	0.80	0.95	·	·	·
Network, 6000 steps*	·	0.84	·	·	·	·
Sentinel-2 only*	0.68	0.74	·	·	·	·
Boosting + Sentinel-2*	0.97	**0.85**	·	·	·	·
NAIP only*	0.62	0.67	·	·	·	·
Boosting + NAIP*	0.97	0.84	·	·	·	·

well (interpolation 0.97, boosting 0.97, the detector 0.95, slope 0.90), and the real USGS 1/3 arc second product behaves almost like the 10 m block mean (interpolation 0.96). At 30 m every elevation layer loses skill. With the settings of our earlier benchmark, interpolation is best (0.82), boosting (0.81) and the network (0.80) follow, and the detector (0.74) beats only slope (0.69) and the aspect rule (0.61). The learned layers were held back by those settings: learned super resolution of the 30 m surface reaches 0.84 and the network trained four times longer 0.84, the best elevation layers at 30 m, and boosting with five more features, among them the interpolated slope, 0.83, which supports the idea that interpolation helps through local curvature. Tuning boosting by nested leave one site out changes nothing (0.81). The information is not gone at 10 m: the largest of the nine 10 m interpolation scores in a 30 m cell reaches 0.96.

Imagery adds what elevation lacks. Boosting on terrain plus Sentinel-2 features reaches 0.85 at 30 m (+0.04 over terrain alone), and with NAIP 0.84; imagery alone is weaker (0.74 and 0.67), and at 10 m it adds nothing (0.97). Super resolution and boosting with Sentinel-2 are above interpolation at all six sites and the detector below it at all six, but with six sites none of these comparisons is significant after Holm correction (sign flip $p = 0.031$, adjusted 0.22).

The real 30 m products are worse than the 30 m block mean: on their own grids the best layer reaches 0.72 on GLO-30 and 0.63 on NASADEM, which predates the lidar by about two decades. Interpolating GLO-30 to the lidar grid, as in our earlier runs, changes little (best 0.72).

The label definition matters less than expected. With the earlier cell label (at least 10 steep posts) the 30 m interpolation scores 0.82 against 0.82; requiring 100 or 300 steep posts raises it to 0.89 and 0.97, because large faces are easier to see; a 30 degree threshold changes little (0.82). The labels are spatially correlated over 30 to 870 m (the largest at Dumont, whose window holds rocky hills); with blocks of 870 m instead of 300 m the per site intervals widen, but slope and the detector stay significantly below interpolation at all six sites.

B. How fine a map must be

Fig. 1a shows how detection falls with grid spacing. Every layer is almost perfect at 2 m (interpolation 1.00) and still good at 10 m (0.97); between 10 and 30 m the AUC of interpolation falls to 0.82 and at 60 m to 0.69, while slope and the detector fall faster. Fig. 1b explains the fall by face size. At every spacing, faces longer than about four grid cells are found almost always (boosting at 30 m: 96 % of faces 4 to 8 cells long), and faces shorter than half a cell are found far less often (60 % at 30 m, 46 % at 60 m). Most faces in these fields are short at 30 m: 75 % of the patches are shorter than one 30 m cell.

C. Planning

Fig. 2 shows every operating point and Table IV the ones chosen on the other five sites. With the posterior vehicle the gate routes at most 34 % of pairs at 10 m and 42 % at 30 m, whatever the layer, because the slip model is uncertain on most dune slopes; most refusals are at an endpoint (76 % of the unrouted pairs for boosting at 30 m and $\tau = 0.2$). The weight policy routes almost every pair and is chosen for every layer on the block means; on 10 m maps the chosen weight is the largest in the grid ($w = 100$), so stronger weights may help further.

On 10 m maps every layer helps. Mission success rises from 17 % with the slip model alone to 64 % with interpolation (+47 points, two stage interval +28 to +64, positive at all six sites), 59 % with boosting and 50 % with the detector. On 30 m maps the gain is small: interpolation adds +5 points (interval +2 to +8) and boosting with Sentinel-2 +5 (interval +1 to +9) to 8 %, and no layer computed from GLO-30 has an interval above

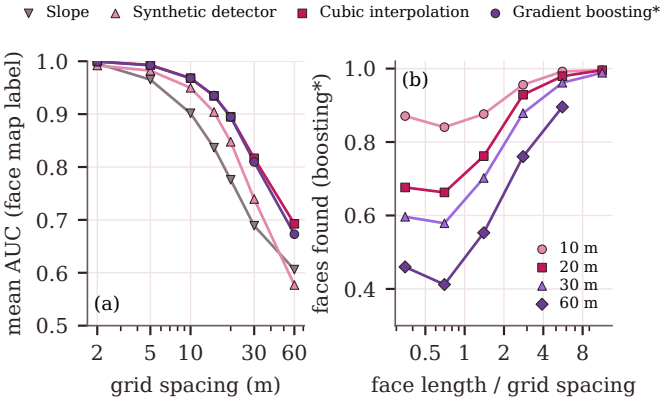


Fig. 1. Resolution and scale (lidar block means, face map label). (a) Mean AUC over six sites against the grid spacing. (b) Share of face patches found by boosting (a cell holding the patch is flagged at the threshold that flags 10 % of face free cells) against the patch length divided by the grid spacing, for four spacings; bins with fewer than 50 patches are omitted. *Trained leave one site out.

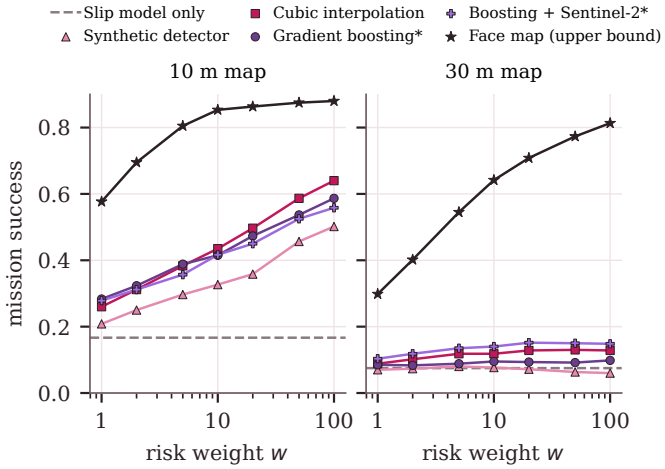


Fig. 2. Mission success of the weight policy against the risk weight w for each hazard layer (E1 endpoints, posterior vehicle, 600 pairs). Dashed: the slip model alone. The face map is an upper bound that needs face information from fine data. *Trained leave one site out.

zero. Endpoints chosen without the truth (E2) lower every rate, for example the slip model alone to 2 % at 30 m, but do not change the comparison.

The face map, an upper bound that assumes face information from fine data, reaches 81 % on the 30 m grid. Degraded face maps show how good that information must be (Table IV, lower part). With the weight fixed at $w = 5$ the face map gives 55 % at 30 m; displaced by 5 m it still gives 50 %, by 10 m 43 % and by 30 m 11 %. Face maps built with the same 28 degree rule from simulated 2 m and 5 m surveys give only 23 % and 8 %: averaging to 2 m already flattens part of the faces below the threshold, and because any contact with a patch of 10 m^2 counts, the patches a survey misses are enough to fail most routes. A survey needs its own threshold, which we have not calibrated.

The outcome is robust to the scoring details. Counting a face within one post of the route (a vehicle footprint) raises the unsafe share of slip only routes at 30 m from 92 % to 94 %; 2 % of the unsafe slip only routes touch faces only while descending the fall line, which some vehicles might survive; and the 28 degree descent rule removes at most 0.1 % of the graph edges on the 30 m grid, so it hardly acts there.

D. Onboard sensing

Table V adds onboard sensing. With a 20 m sensing radius the rover reaches the goal without touching a face on 80 % of the pairs when its 30 m prior holds only the slip model, and on 84 % with boosting as prior (+4 points, interval -1 to +12); even the face map prior gives 85 %. Sensing, not the prior, keeps the rover off faces: no route with E1 endpoints touched one. The failures are pairs where sensed faces close every way to the goal (20 % with the slip model prior). A better prior saves replanning rather than distance: the rover replans 29 times per pair with the slip model prior and 8 times with the face map, while the driven length stays near 1.16 times the straight distance. A 10 or 30 m radius gives 76 % and 83 %

TABLE IV
MISSION SUCCESS (%) AT THE OPERATING POINT CHOSEN ON THE OTHER FIVE SITES (GATE THRESHOLD OR WEIGHT), 100 PAIRS PER SITE, POSTERIOR VEHICLE. E1: ENDPOINTS CHOSEN WITH THE TRUTH; E2: CHOSEN FROM THE COARSE MAP ONLY. †TWO STAGE BOOTSTRAP INTERVAL OF THE GAIN OVER THE SLIP MODEL ENTIRELY ABOVE ZERO. DEGRADED FACE MAPS: WEIGHT POLICY, $w = 5$. *TRAINED LEAVE ONE SITE OUT.

	I10		I30		G30
Layer	E1	E2	E1	E2	E1
Slip model only	17	9	8	2	6
Slope	35†	28†	6	2	6
Interpolation	64†	53†	12†	5	6
Detector	50†	33†	7	4	5
Detector, recalibrated	52†	37†	5	2	6
Boosting*	59†	47†	8	4	7
Network*	50†	40†	6	3	7
Boosting + Sentinel-2*	56†	46†	13†	7†	·
Face map (upper bound)	84†	80†	81†	68†	80†
Face map, $w = 5$	80	71	55	40	·
Face map 5 m off	67	56	50	39	·
Face map 10 m off	36	26	43	32	·
Face map 30 m off	19	10	11	5	·
Face map from 2 m survey	29	19	23	14	·
Face map from 5 m survey	17	9	8	4	·

with the slip model prior, and E2 endpoints give the same pattern at a lower level (67 % and 71 %).

V. DISCUSSION AND LIMITATIONS

What the results mean. A 30 m elevation model keeps enough of the dune shape to rank face cells better than chance, but not enough to plan around faces: the best 30 m layer adds a few points of mission success, while the same layers on a 10 m map add +47 points. The scaling curve (Fig. 1b) explains why: a face is found reliably once it spans about four grid cells, and most faces in these fields are shorter than one 30 m cell. Better learning helps detection a little (a longer trained network reaches 0.84 against 0.82 for interpolation), but the shape of the curve, not the choice of model, sets the

TABLE V
CLOSED LOOP WITH ONBOARD SENSING (RADIUS R), 20 PAIRS PER SITE, E1 ENDPOINTS, POSTERIOR VEHICLE: SUCCESS (%; GOAL REACHED, NO FACE CONTACT) AND REPLANS PER PAIR (RP). MAIN RUNS $R = 20$ M; I30 ALSO WITH $R = 10$ AND 30 M.

Prior layer	I10, R 20 S	I10, R 20 Rp	I30, R 20 S	I30, R 20 Rp	I30, S R 10	I30, S R 30
Slip model only	78	11	80	29	76	83
Detector	81	5	85	25	82	85
Interpolation	80	4	82	20	81	87
Boosting*	80	4	84	21	81	88
Face map	80	1	85	8	82	86

limit. Imagery gave the largest gain in 30 m detection, which matches the experience of dune mapping from imagery [21]; in planning its gain was the same size as the interpolation's. The face map results put the remaining gap on information, not on the planner: a face map displaced by 5 m keeps most of its benefit on the 30 m grid, but the benefit falls quickly beyond that (Table IV), so a drone survey must be georeferenced to a few metres and its face threshold calibrated for its own resolution.

Arabian dunes. We did not test Arabian fields. The scaling curve suggests what to expect: a slip face 100 m high at 32 degrees spans about 160 m, five 30 m cells, which GLO-30 should resolve, while the smaller faces on the flanks of megadunes and on the linear dunes of the Rub' al Khali would remain hidden. The curve was measured on six North American fields and TanDEM-X, the source of GLO-30, performs worse over sandy deserts [24], so this is an expectation to test, not a result.

Limitations. The truth is steep ground: rock, vehicle ruts and vegetated mounds also exceed 28 degrees, and the manual audit of a stratified sample of 600 patches is not finished, so label precision is unknown. The windows were chosen by us and used in earlier tests of the detector, and the prediction log is kept in our own repository rather than a public registry. The vehicle model is a posterior fitted to published slip curves of other vehicles; in our own leave one study out validation no slip model family transferred between published studies (median held out slip error 0.20 to 0.22), so absolute route outcomes are uncertain. No rover was driven and no physics was simulated; the closed loop assumes perfect sensing of faces within the radius. Any contact with a face patch counts as a failure, which is strict, although the footprint and descent variants change the picture little. Six sites limit the training data of the learned layers and the power of site level tests (the smallest attainable two sided p is 1/32).

VI. CONCLUSION

A 30 m elevation model ranks slip faces better than chance but cannot keep a small rover off them. On six lidar dune fields no hazard layer computed from such a map, learned or not, with or without imagery, added more than a few points of mission success, while the same layers on a 10 m map raised it from 17 % to as much as 64 %. Faces shorter than about four grid cells are the problem, so the survey resolution a rover needs depends on the size of the faces in its field. In simulation, onboard sensing within 20 m kept every route between gentle endpoints off faces, and a better prior mainly saved replanning. The next steps are the manual audit of the face labels, the same benchmark on Arabian dune fields with a drone survey, and a rover whose slip has been measured on the same sand.

DATA AND CODE

USGS 3DEP lidar and DEMs, NASADEM and NAIP are in the public domain; Sentinel-2 data are provided under the Copernicus free licence; Copernicus GLO-30 is used under its licence (produced using Copernicus WorldDEM-30 © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018, provided under COPERNICUS by the European Union and ESA; all rights reserved). The code, result files, prediction log and the scripts that regenerate every number and figure of this paper and its supplement are at https://github.com/hudasol/dune-rover (stages 25 to 28).

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